



LALLY SCHOOL
OF MANAGEMENT

SYLLABUS

Web Systems Development Fall 2026

ITWS 2110/CSCI 2960

Room Location: Lally102

Class Time: Tuesday/Friday 12:00 PM – 1:50 PM (01); 2:00 PM – 3:50 PM (02);
RPILMS

INSTRUCTOR (REQUIRED)

Instructor Name: Jason Kuruzovich

Office Location: Pitt 2206

Tel. No.: 518-698-9910

Email Address: kuruzj@rpi.edu*

Office Hours: Wed 1:00 - 3:00 (in person); Teams by appointment

Book online appointment: <https://bit.ly/jason-rpi>

COURSE DESCRIPTION:

This course involves a study of the methods used to extract and deliver dynamic information on the World Wide Web. The course uses a hands-on approach in which students actively develop Web-based software systems. Additional topics include installation, configuration, and management of Web servers. Students are required to have access to a PC on which they can install software such as a Web server and various programming environments.

REQUIRED TEXT(S):

The reading materials will be tailored to provide the needed conceptual background for the various technologies examined. Links to the required readings will be available via the LMS and/or course Github.

STUDENT LEARNING OUTCOMES

By the end of the course, students will be able to:

1. Explain how a full-stack web application works—from the browser to the server, database, and network.
2. Select and justify appropriate technologies, architectures, and design patterns.
3. Build effective front-end experiences using HTML, CSS, JavaScript, and progressive enhancement.
4. Develop, integrate, and troubleshoot application back ends and web services.

5. Identify common web security risks and apply protections such as input validation, secure authentication, authorization, and safe data handling.
6. Explain how AI is used in web development and applications, critically evaluate its outputs, and recognize its capabilities, limitations, and security implications.
7. Plan, build, test, deploy, and present a complete web application using modern development practices.

COURSE ASSESSMENT MEASURES

<u>Assessment</u>	<u>Description</u>	<u>Learning Outcome #s</u>
Labs/Homeworks	Each <i>individual</i> will submit In-class lab labs and homeworks.	1-9
Quizzes	Each <i>individual</i> will take two quizzes, one around the midpoint of the semester and the other after all lectures have been given.	1, 2, 7
Project	Each <i>group</i> will design, implement, iterate on, and present a semester-long web application project. The group will include overall and individual contributions.	1, 2, 3, 4, 5

GRADING CRITERIA

*Important note: You must have a passing average on **each** component (labs/homework, group project, quizzes, and participation) to pass the course. (Quizzes exempted.)*

Grading Schedule

SUMMARY OF COURSE ASSESSMENT MEASURES		Grades	
Assessment	Weight		
Lab/Homework Assignments	30%	>93	A
Semester Group Project	30%	90-92	A-
– Overall System Architecture & Performance	15%	87-89	B+
– Individual Contribution & Professional Practice*	15%	83-86	B
Quizzes	30%	80-82	B-
Participation	10%	77-79	C+
Total	100%	73-76	C
		70-72	C-
		66-69	D
		0-65	F

*Will incorporate peer feedback, project commits, instructor observation, individual summary of contribution.	
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ATTENDANCE POLICY

Attendance is mandatory. Any absence must be communicated to the instructor in advance. Attendance is a part of graded component of this course. **Student is responsible for signing the attendance sheet at the beginning of the class.**

Late Policy

Late submissions are accepted for **one week** following the due date with a **10% grade reduction**. Submissions more than one week late receive a **30% reduction**.

No submissions are accepted after **Friday, December 11** — the last day of classes. The final project report is due during the exam period and is not subject to this cutoff.

In-class lab assessments cannot be submitted late. A quiz or presentation missed without a prior approved excuse is graded zero. Excused absences are arranged in advance; see the attendance policy.

Extensions for documented circumstances are granted freely **in advance**. Ask before the deadline, not after.

OTHER COURSE POLICIES

Students must follow all institute safety and health protocols at all times throughout the semester. There will be no exception to these requirements at any time.

ACADEMIC INTEGRITY

Student-teacher relationships are built on trust. For example, students must trust that teachers have made appropriate decisions about the structure and content of the courses they teach, and teachers must trust that the assignments that students turn in are their own. Acts that violate this trust undermine the educational process.

The Rensselaer Handbook of Student Rights and Responsibilities and the Graduate Student Supplement (For 6000 level and above courses) define various forms of Academic Dishonesty and you should make yourself familiar with these. In this class, all assignments that are turned in for a grade must represent the student's own work. In cases where help was received, or teamwork was allowed, a notation on the assignment should indicate your collaboration. Submission of any assignment that is in violation of this policy will result in (1) an academic (grade) penalty and (2) reporting to Lally's Associate Dean of Academic Affairs and either the Dean of Students (for Undergraduates) or the Dean of Graduate Education (for Graduate students).

In this course, the academic penalty for a first offense is *loss of a letter grade for the course*. A second offense will result in failure of the course as noted in Lally's Three Strikes Policy.

If you have any questions concerning this policy before submitting an assignment, please ask for clarification.

ACADEMIC ACCOMMODATIONS

Rensselaer Polytechnic Institute strives to make all learning experiences as accessible as possible. If you anticipate or experience academic barriers based on a disability, please let me know immediately so that we can discuss your options.

To establish reasonable accommodations, please register with The Office of Disability Services for Students (<mailto:dss@rpi.edu>; 518-276-8197; 4226 Academy Hall). After registration, make arrangements with me as soon as possible to discuss your accommodations so that they may be implemented in a timely fashion."

Disability Services

Rensselaer Polytechnic Institute strives to make all learning experiences as accessible as possible. If you anticipate or experience academic barriers based on a disability, please reach out to Disability Services immediately to discuss your options. To establish reasonable accommodations, please register with The Office of Disability Services for Students as soon as possible to discuss your accommodations so that they may be implemented in a timely fashion. DSS contact information: dss@rpi.edu; +1-518-276-8197; 4226 Academy Hall.

Support Services

RPIInfo - contains various resource links for students, academic resources, support services, and safety & emergency preparedness.	Accommodations	518.276.2231	https://studenthealth.rpi.edu/list-services/disability-student-services
Academic Assistance	Executive Functioning		
Tutoring	Success Coaching		
Success Coaching		518.276.6269	https://success.studentlife.rpi.edu/
Student Health & Wellness	Counseling Center	518.276.6479	https://studenthealth.rpi.edu/list-services/counseling-center
Mantra Health (after hours counseling & wellness support)		518-276-8888	http://mantrahealth.com/RPI
Student Health Center		518.276.6287	https://studenthealth.rpi.edu/list-services/student-health-center
Student Support Services & Conduct	Dean of Students Office	518.276.6266	https://doso.rpi.edu/
RPI Info	Central Directory for RPI Services		https://info.rpi.edu
DOTCIO (IT Services)	Help Desk	518.276.7777	Submit a Help Desk ticket: https://itssc.rpi.edu/hc/en-us

Tentative Course Calendar

Fall 2026 — ITWS 2110 · Aug 27 – Dec 11, 2026 · Tuesday/Friday · 28 sessions

#	Date	Session	Due 11:59 PM
1	Fri 8/28	Course intro. The modern web application. How grading works: clone, run the suite, red to green	
2	Tue 9/1	Docker and infrastructure as code	In-class
3	Fri 9/4	LAMP stack on Docker; the GitHub workflow — clone, branch, pull request	HW 0: Environment + repo
4	Tue 9/8	Test-driven development — tests as specification; Pest/PHPUnit anatomy	In-class
5	Fri 9/11	Writing tests; CI runs your suite on every push · teams form · last day to add/drop	HW 1: Dockerized LAMP stack
6	Tue 9/15	CSS frameworks I — Bootstrap / Tailwind; utility vs. component approaches	In-class
7	Fri 9/18	CSS frameworks II — responsive layout, design systems; when not to use one	HW 2: Testing infrastructure
8	Tue 9/22	React I — components, props, the render model	In-class
9	Fri 9/25	React II — state, events, forms · project proposal	HW 3: CSS framework build
10	Tue 9/29	React III — lists, lifting state, composition	In-class
11	Fri 10/2	React IV — component testing with Vitest and Testing Library	HW 4: React components
12	Tue 10/6	Application frameworks: Laravel I — what a framework buys you; MVC, routing, Blade	In-class
13	Fri 10/9	Laravel II — controllers, requests, HTTP tests · Quiz 1 review	HW 5: React with component tests
14	Tue 10/13	QUIZ 1 — handwritten, closed-device (covers sessions 1–11)	
—	Fri 10/16	No class — follows a Monday schedule	
15	Tue 10/20	ORM I — Eloquent, migrations, seeders; schema as code	In-class
16	Fri 10/23	ORM II — relationships, eager loading, the N+1 problem; database testing	HW 6: Laravel app (2-wk)
17	Tue 10/27	Authentication, Forms, validation, sessions, flash messages, CSRF tokens	In-class
18	Fri 10/30	Authentication — authn vs. authz, password storage, Breeze, middleware, policies	HW 7: Data model + ORM
19	Tue 11/3	SECURITY — OWASP Top 10:2025; injection, XSS, CSRF; headers, TLS, secrets, dependency and supply-chain risk · project midpoint · last day to drop with W	In-class
20	Fri 11/6	Security Lab	HW 8: Forms + authentication
21	Tue 11/10	APIs — JSON, REST, resource design, versioning, documentation, contract testing; authenticating an API (Sanctum) vs. a session	In-class
22	Fri 11/13	API Lab — wire your React front end to your Laravel API	HW 9: Security hardening + findings

#	Date	Session	Due 11:59 PM
23	Tue 11/17	Cloud deployment — build, ship, run; environments, secrets, migrations in production	In-class
24	Fri 11/20	Deployment Lab	HW 10: JSON API + React consuming it
—	11/23–27	<i>Thanksgiving Break — no classes</i>	
25	Tue 12/1	Performance — caching layers, revisiting N+1, load testing, measuring before/after	In-class
26	Fri 12/4	QUIZ 2 — handwritten, closed-device (covers sessions 15–24)	
27	Tue 12/8	Final presentations prep day and wrap-up.	HW 11: Deployment
28	Fri 12/11	Final presentations	Presentation
—	12/16–22	<i>Final project report due — exam period</i>	

Dates verified against the RPI registrar academic calendar. Schedule is tentative; the LMS is authoritative.